Short Report

Single centrifugation of cerebrospinal fluid in a sealed Pasteur pipette for simple, rapid and sensitive detection of trypanosomes

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Keywords: trypanosomiasis, *Trypanosoma brucei gambiense*, cerebrospinal fluid, diagnosis, centrifugation, methodology

Treatment of human African trypanosomiasis (HAT) due to *Trypanosoma brucei gambiense* requires distinction between haemo-lymphatic and meningo-encephalitic stages. The presence of trypanosomes in the cerebrospinal fluid (CSF) is not only evidence supporting diagnosis, but also indicates central nervous system invasion (WHO, 1998). When their number in CSF is high, trypanosomes are detectable during the cell count. In the case of low trypanosome number, detection is facilitated by centrifugation of CSF and microscope examination of the sediment (LAVERAN & MESNIL, 1912). CATTAND *et al.* (1988) improved the sensitivity of trypanosome detection by introducing a double centrifugation (DC) technique: after the first centrifugation, the sediment is taken up in micro-haematocrit tubes which are centrifuged and examined for trypanosomes at the bottom of the tubes. The DC technique is not widely practised owing to the number of manipulations and the need for 2 centrifuges. Here we describe a modified single-centrifugation (MSC) technique as an alternative to DC, which is compared with other WHO therapeutic parameters (WHO, 1998).

The data were collected at the Projet de Recherches Cliniques sur la Trypanosomiase Daloa, Côte d'Ivoire. For routine diagnosis and stage determination of HAT, 8 mL of CSF were obtained from 52 card agglutination test for trypanosomiasis (CATT)-positive persons by lumbar puncture. According to WHO (1998), the following examinations were performed: parasite detection, cell count and protein concentration. Trypanosomes were searched for in blood by the mini-anion exchange centrifugation technique (mAECT), quantitative buffy coat and kit for in-vitro isolation, and in lymph by examination of the lymph-node aspirate. Presence of trypanosomes in CSF was examined by DC of 6 mL of CSF and in parallel by MSC. For MSC, a sealed Pasteur pipette was prepared as for mAECT (LUMSDEN *et al.*, 1979). The pipette was filled with 2 mL of CSF and centrifuged (10 min, 600 ***g***). Using the mAECT viewing-chamber, the sediment in the bottom of the tube was examined by microscopy for the presence of trypanosomes. The cell number in CSF was counted using a Malassez counting chamber and the CSF protein concentration was assayed with the Coomassie brilliant blue method. The different methods were compared using McNemar χ^2 for $\alpha = 0{\cdot}05$.

Combining all techniques, trypanosomes were detected in 42 (80·8%) of the 52 patients. According to WHO criteria (WHO, 1998), 35 of these patients (83·3%) were in the meningo-encephalitic stage. Trypanosomes were detected in 34 CSF samples; the DC and MSC techniques were positive in, respectively, 28 and 33 patients (Table). MSC detected trypanosomes in 6 samples negative with DC (with 6, 8, 10, 20, 194 and 338 cells/µL in CSF), whereas DC detected trypanosomes in only 1 MSC-negative sample (with 726 cells/µL). Although MSC appeared more sensitive, the difference of sensitivity between MSC and DC was not statistically significant ($\chi^2 = 2{\cdot}29$; $P > 0{\cdot}05$). There was no significant difference between the MSC technique and the white cell count ($\chi^2 = 0{\cdot}50$; $P > 0{\cdot}05$) or total protein concentration in CSF ($\chi^2 = 0{\cdot}57$; $P > 0{\cdot}05$) as criteria for stage determination.

We conclude that MSC is rapid, simple and sensitive for diagnosis and stage determination of HAT. The test can be performed in 10 min. Trypanosomes can be detected in the bottom of the sealed Pasteur pipette, avoiding the need to transfer the sediment on to a slide. MSC is less elaborate than DC since only 1 centrifugation step is needed and is cheaper since only a sealed pipette and 1 centrifuge are necessary. MSC is also recommended for post-treatment follow-up.

Table. Comparison of the modified single-centrifugation technique (MSC) with double centrifugation (DC), cerebrospinal fluid (CSF) cell count and protein concentration as aids for diagnosis of African trypanosomiasis

	MSC		
Other techniques	Negative	Positive	Total
DC			
Negative	8	6	14
Positive	1	27	28
Total	9	33	42
CSF cell count			
≤5 cells/mm³	7	0	7
>5 cells/mm³	2	33	35
Total	9	33	42
CSF protein content			
≤370 mg/L	7	5	12
>370 mg/L	2	28	30
Total	9	33	42

Acknowledgements

This study received financial support from the Ministère de la Santé, Côte d'Ivoire, and the General Administration for Development Co-operation, Belgium. We thank Mr Konan Konan Alexandre, Dion Ladaidé, Gabin Ferdinand, Auguste Bernard and Blé Léopold for their technical assistance.

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Received 3 September 1999; revised 17 December 1999; accepted for publication 17 December 1999

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Key: CSF, Tdae, sal, Tb, Tbg, diagnosis, centrif, method, ss.

MIÉZAN-'00 8739

Short Report

Hypothalamic neurocysticercosis as a possible cause of obesity

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Keywords: cysticercosis, *Taenia solium*, hypothalamus, obesity, case reports, Brazil

We describe 2 cases of hypothalamic neurocysticercosis (NCC) in association with obesity. A possible central mechanism affecting regulation of alimentary ingestion secondary to NCC is suggested. We believe that such association has not been previously reported.

Case reports

Case 1

A 48-year-old woman was admitted to the hospital with severe abdominal pain. Physical examination showed an obese patient, bodyweight 80 kg, height 1·55 m, body mass index (BMI) 33·7 kg/m^2. No data regarding any clinical history of NCC were found in the medical records. She evolved to death and autopsy revealed chronic pancreatitis and gallbladder lithiasis. The brain weighed 1180 kg, presenting a cysticercus located in the anterior hypothalamus, with intense chronic inflammatory reaction and necrosis around it.

Case 2

A 62-year-old woman, bodyweight 90 kg, height 1·53 m, BMI 38·5 kg/m^2, with clinical history of diabetes mellitus, was admitted to the hospital presenting dyspnoea, sweating, and cold extremities. Physical examination revealed abdominal distension, and a periumbilical mass. She evolved to death and an autopsy was performed showing a gallbladder adenocarcinoma infiltrating the right hepatic lobe. A cysticercus of 1-cm diameter was found compressing the third ventricle with partial hypothalamic atrophy (Figure). There were amyloid corpuscles, perivascular lymphocytic infiltrate, and neovessel formation around the cysticercus.

Comments

NCC is a central nervous system infection caused by the presence of *Taenia* larvae and is responsible for a variety of neurological manifestations (BUITRAGO *et al.*, 1995). We describe 2 cases of NCC with mechanical and inflammatory lesions in the anterior hypothalamus probably associated with the patients' obesity. NCC may have contributed to development of obesity, probably associated with hypothalamic hyperphagia syndrome. The hypothalamic medial ventral nucleus (MVN) has been related to obesity in rats (GOLD *et al.*, 1977). The MVN is considered an integrated centre forming part of a wide neuronal circuit which influences control of food and water intake (GOLD *et al.*, 1977). There is evidence that tissue immediately beside or rostral to the MVN may be important for producing the effect of stimulating hunger in rats rather than inhibiting satiation (LEIBOWITZ *et al.*, 1981).

The hyperphagia and weight gain observed after an MVN lesion are attributed in part to hyperinsulinaemia due to disturbance in the autonomic control of pancreatic islets (INOUE *et al.*, 1983). The control of food intake must furthermore involve the influence of hypothalamic receptors on the hormone called leptin, which would give information on the status of body adipose tissue (FEI *et al.*, 1997). There are case reports of hypothalamic craniopharyngioma in humans associated with obesity, which would indicate a disturbance in the hypothalamic circuit for control of food intake and/or sensitivity to leptin (ROTH *et al.*, 1998).

The hypothalamic NCC with inflammatory and mechanical lesions in the anterior hypothalamus described here suggests an anatomical basis for the patients' obesity, probably due to hypothalamic hyperphagia syndrome.

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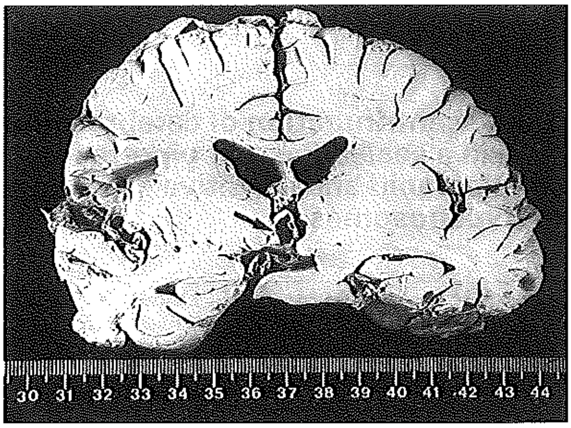

Figure. Brain of Case 2, gross anatomy, sagittal section showing a cysticercus located in the anterior hypothalamus (arrow).

Acknowledgement

We thank Sandra J. Olson for revising the manuscript.

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Received 8 September 1999; revised 17 January 2000; accepted for publication 18 January 2000